

# Private Credit Underwriting System Explanation

A governed vertical slice from source conflict to reviewed underwriting and an evidence-linked IC memo.

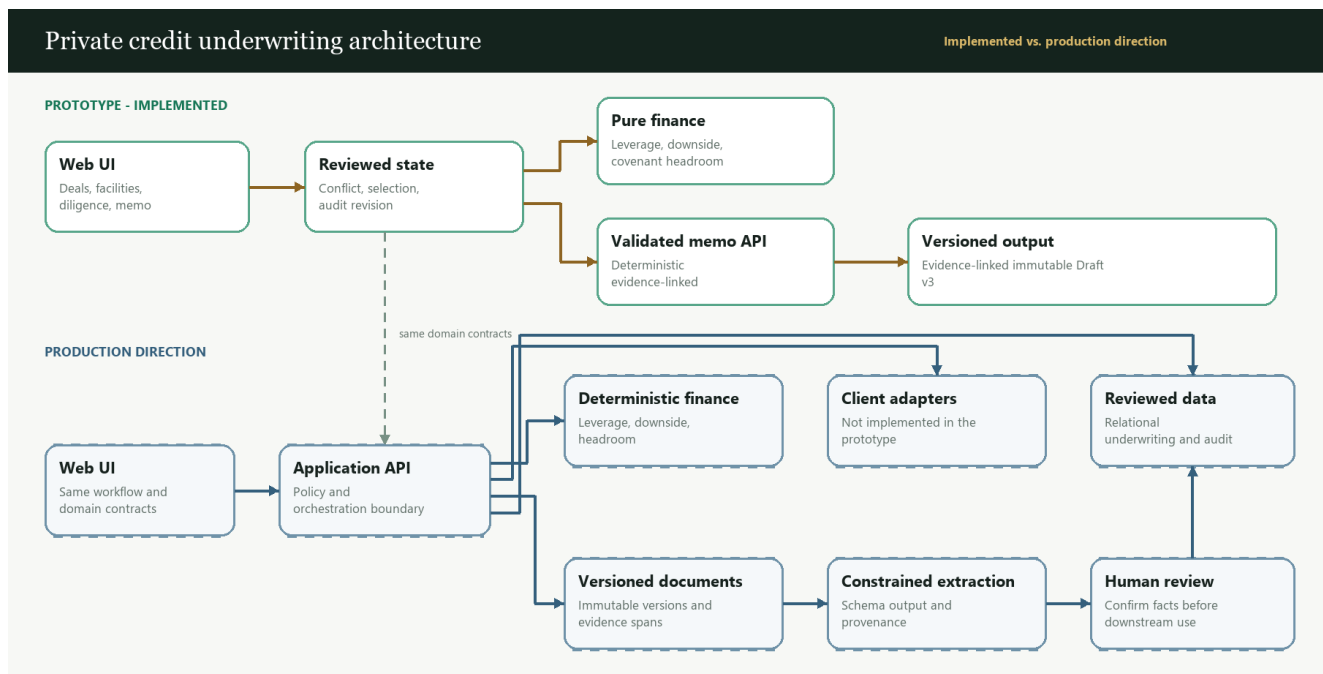
## 1. What I built

I built a private-credit workspace connecting Deals, Securities, Due Diligence, and IC Memo through one Atlas Industrial Services workflow. Management materials assert \$33.0m EBITDA while the QoE supports \$30.0m. Both source assertions remain visible. After reviewing the seeded evidence, the Demo Analyst selects **Use \$30.0m for underwriting** without overwriting either source.

Pure functions update base leverage (4.00x to 4.40x), 20% downside EBITDA (\$26.4m to \$24.0m), downside leverage (5.00x to 5.50x), and headroom to the synthetic 5.75x covenant (0.75x to 0.25x). The human action creates an audit event and makes immutable Draft v2 out of date. A validated endpoint deterministically composes evidence-linked Draft v3; no live model call is made.

The product supports, rather than replaces, investment judgment. Resolving a material conflict once and propagating the reviewed value through facility metrics and the IC memo reduces re-keying, improves consistency, and returns reviewer time.

## 2. Architecture



The prototype is a small TypeScript monolith: React UI, memory-only reducer state, pure finance functions, and one validated memo endpoint. In production, immutable document versions feed constrained extraction, provenance, and human review. Reviewed structured data is the system of record; the memo is a versioned view. Client-system adapters are production direction only and are not implemented or simulated here.

3. Data model

- **Deal-related records** include **Security**, **Document**, **DiligenceItem**, **Finding**, **MemoVersion**, and **AuditEvent**, linked by deal ID. The \$120m term loan and \$25m revolver are separate Securities.
- **DocumentVersion** is immutable. Evidence identifies its document version, section, and excerpt; no page numbers are fabricated.
- **SourceAssertion** records value, unit, provenance, and evidence. Conflicts coexist; separate assertions also source net debt and covenant inputs, and composition rejects drift from cited observations. **UnderwritingSelection** is canonical reviewed EBITDA state and advances revision without rewriting sources.
- **Finding** is the risk/mitigant record. It can reference multiple Evidence records. The UI projects QoE review from the canonical selection, so it becomes Confirmed while remaining High and Open.
- **MemoVersion** records its selection ID and underwriting revision; MemoClaims link to Evidence. AuditEvents capture before/after selection state. **semanticOrigin** distinguishes source observation, system composition, and human/model states; **materializationSource** separately discloses static demo data versus runtime code.

4. Tradeoffs and prototype truth ledger

| CLASSIFICATION   | INCLUDED SCOPE                                                                                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REAL             | Navigation; conflict/evidence review; human selection; finance/state propagation; audit; memo staleness, immutable comparison/composition, evidence links; reset.  |
| MOCKED           | Signed-in Demo Analyst identity and review rights.                                                                                                                 |
| STATIC DEMO DATA | Three deals, excerpts, assertions, findings, securities, and most Draft v2 prose.                                                                                  |
| NOT IMPLEMENTED  | Live ingestion/model calls, persistence, real auth/permissions, client-system adapters/connectors, full covenant parsing, multitenancy, and production operations. |

State intentionally resets on refresh. I simplified ingestion and persistence with seeded fixtures and memory-only state. I mocked only the signed-in analyst. I ignored post-close monitoring and servicing because they do not prove the underwriting-to-IC path.

For this publicly accessible take-home demo, I deliberately used a **deterministic, rule-based memo composer instead of connecting the application to a live LLM API**. This keeps the walkthrough reproducible and avoids exposing an unauthenticated generation endpoint to abuse and uncontrolled provider cost. In production, the same validated service boundary could invoke a schema-constrained model behind server-side authentication, authorization, quotas, and audit, while finance calculations and investment approval remain deterministic and human-controlled.

5. What I would add with two more days

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DAY 1 | <p><b>Constrained ingestion + durable reviewed state</b></p> <p><b>Morning (1/2 day):</b> Accept text-native PDFs, create immutable document versions with page- and section-level evidence, and route scanned or unsupported files to Needs Review rather than guessing.</p> <p><b>Afternoon (1/2 day):</b> Persist assertions, selections, findings, evidence links, append-only audit events, and memo versions.</p> <p><b>DONE MEANS:</b> A refresh no longer resets the analyst's selection, and the \$33.0m/\$30.0m conflict remains fully traceable.</p>                                                                   |
| DAY 2 | <p><b>Evaluated model path + operational guardrails</b></p> <p><b>Morning (1/2 day):</b> Add one schema-constrained extraction and drafting path behind the existing service boundary, with golden fixtures, evidence-reference validation, and rejection of unsupported claims.</p> <p><b>Afternoon (1/2 day):</b> Add provider timeout/retry handling, deterministic fallback, and reviewer accept/edit/reject telemetry.</p> <p><b>DONE MEANS:</b> The Atlas workflow survives a provider failure, every generated claim stays evidence-linked, and the model never calculates finance or approves an investment decision.</p> |